Propositional Logic Series
Episode 2: Formal Definitions and Notation

Purpose

This short sheet collects the basic definitions and notation used in Episode 2. The aim is not to replace the video, but to fix the terminology before working on the exercises.

1. Truth-Apt Expressions and Propositions

Definition. A truth-apt expression is an expression that can be true or false.
Definition. In this series, a proposition is a truth-apt declarative expression.
Examples.

- "24 is divisible by 4" is truth-apt.
- "Every prime number greater than 2 is odd" is truth-apt.
- "Is 24 divisible by 4?" is not truth-apt in the strict logical sense. It is a question.
- "Prove the theorem" is not truth-apt in the strict logical sense. It is a command.
- "Awesome!" is not truth-apt in the strict logical sense. It is an exclamation.
- "Let us go to the cinema tonight" is not truth-apt in the strict logical sense. It is a direct request or suggestion.

2. Argument, Premise, and Conclusion

Definition. An argument is an expressed structure of reasoning in which some truth-apt expressions are presented as supporting another truth-apt expression.
Definition. A premise in an argument is a truth-apt expression used as a reason or ground in the reasoning.
Definition. A conclusion in an argument is the truth-apt expression whose truth or acceptability the reasoning is intended to establish or support.
Example. Consider the following argument:

> If an integer is divisible by 6, then it is divisible by 3.
> 120 is divisible by 6.
> Therefore, 120 is divisible by 3.

Here, the first two expressions are premises, and the last expression is the conclusion.

3. Standard Form

Putting an argument into standard form means writing its premises first and its conclusion last.
Standard form.

> Premise 1: If an integer is divisible by 6, then it is divisible by 3.
> Premise 2: 120 is divisible by 6.
> Conclusion: 120 is divisible by 3.

Standard form does not change the argument. It makes the structure of the argument explicit.

4. Indicator Words

Some words help us identify the role of a statement in an argument.
Conclusion indicators.

- therefore
- thus
- hence
- so
- consequently
- it follows that

Premise indicators.

- because
- since
- given that
- assuming that
- for the reason that

Indicator words are helpful, but they are not definitions. The role of an expression depends on how it functions in context.

5. Argument, Explanation, and Mere List

Definition. A passage is a mere list if it presents truth-apt expressions without presenting some of them as support for another.
Definition. An explanation gives a reason why an already accepted, observed, or assumed fact is the case.
Working distinction.

- An argument tries to show that a conclusion is true or supported.
- An explanation tries to show why something already accepted or observed is the case.

Example.

> The ground is wet because it rained last night.

If the wet ground is already observed, this may function as an explanation. If the wetness of the ground is being inferred from other information, it may function as an argument.

6. Deductive and Inductive Arguments

Definition. A deductive argument is an argument that presents its conclusion as something that cannot be false if all its premises are true.
Definition. An inductive argument is an argument that presents its conclusion as supported, made probable, or made reasonable by its premises, but not as forced by them.
Deductive example.

> If an integer is divisible by 6, then it is divisible by 3.
> 120 is divisible by 6.
> Therefore, 120 is divisible by 3.

Inductive example.

> 10 is divisible by 5 and by 2.
> 20 is divisible by 5 and by 2.
> 30 is divisible by 5 and by 2.
> Therefore, every natural number divisible by 5 is divisible by 2.

The conclusion is false. For example, 15 is divisible by 5 but not by 2.

7. Validity Preview

Definition. A deductive argument is valid if and only if there is no case in which all premises are true and the conclusion is false.
Definition. A deductive argument is invalid if and only if there is at least one case in which all premises are true and the conclusion is false. This will be the main topic of the next video.

8. Inference, Consequence, and Entailment

Inference often refers to the act or process of drawing a conclusion. **Consequence** or **entailment** refers to the logical relation between premises and a conclusion. It is better to say:

- We infer a conclusion from premises.
- A conclusion follows from premises.
- The premises entail the conclusion.

Avoid saying that "the premises infer the conclusion." Inference is something reasoners do; entailment is a relation between propositions.

Sources and Note

The terminology and conceptual background in this sheet are informed by P. D. Magnus, Tim Button, Robert Trueman, and Richard Zach, *forall x: Calgary: An Introduction to Formal Logic*, with contributions by J. Robert Loftis and Aaron Thomas-Bolduc.

Official page: `https://forallx.openlogicproject.org/`

This handout is an original teaching resource prepared for the Propositional Logic Series. It is not a reading, summary, or reproduction of the textbook.